

Exploratory Data Analysis

Zuil Pirola

Data-Analytic Thinking

Data-analytic thinking is a **systematic approach** to **problem-solving** that places data at the forefront of knowledge extraction and decision-making.

<https://medium.com/data-beyond/what-is-data-analytics-thinking-and-why-data-leaders-must-have-it-ebcb8285a5d6>

Data-Analytic Thinking: Historical Roots

History of Analytics

- Data-analytic thinking existed long **before computers**.
- Humans have always collected **observations** to support decisions.

Ex: "He is dead?", "This is poisonous."

- Early analytics emerged from survival, trade, health, and governance needs.
- The **tools changed** — the **mindset persisted**.

Business Analytics and Data-Analytic Thinking

Business Analytics vs Data-Analytic Thinking

- Data-Analytic Thinking: Data-driven mindset for **understanding problems and making decisions**.
- Business Analytics: Application of data analysis methods to **improve business** outcomes.
- Relationship: Analytical thinking guides how data is interpreted; **business analytics applies it in practice**.
- Key Idea: ***Thinking with data enables better business decisions.***

Analytics in Early Barter Economies

Business Analytics Before Writing Tools

- Early **trade required tracking value**, exchanges, and trust.
- Without paper or computers, people likely used:
 - **Marks on cave walls or bones**
 - Stones, wood tallies, or symbolic tokens
- These represent **primitive data recording systems**.
- Even then, decisions depended on **accumulated observations**.

First evidence of **business analytics in human history**.

19th Century Data Thinking in Public Health

Data Saving Lives Before Computers

Two landmark examples:

- **Ignaz Semmelweis (1840s):**

- Demonstrated **handwashing reduced maternal mortality**.
- Used observational hospital data.
- Faced resistance despite evidence.

- **Rise of statistical thinking in medicine and demography.**

Data analysis began influencing scientific and social decisions.

Semmelweis and the Handwashing Hypothesis (1840s)

Ignaz Semmelweis: Early Data-Analytic Thinking

- **Hungarian physician** working at Vienna General Hospital.
- Investigated high maternal mortality from **puerperal fever**.
- Observed higher deaths in the physicians' ward **vs** midwives' ward.
- Began systematic **comparison** of hospital data.

Key idea: Observation + data comparison led to hypothesis formation.

The Experiment and Data Evidence

Testing the Handwashing Intervention

- Hypothesis: infections linked to cadaveric particles from autopsies.
- Introduced **handwashing with chlorinated lime solution (1847)**.
- **Mortality dropped dramatically (~10–18% → ~1–2%)**.
- Clear **quantitative evidence** supporting intervention.

Example of:

- Early **experimental** intervention
- **Data-driven** decision making
- Evidence **before germ theory** was established

Lessons for Data Analytics and EDA

Why Semmelweis Matters Today

- **Data can reveal hidden causes** of real-world problems.
- **Correlation** can guide investigation and action.
- **Resistance to data-driven** change is common.
- Strong example of exploratory analysis informing decisions.

Core takeaway:

Data-analytic thinking can save lives — even before full scientific explanation exists.

John Snow and the Cholera Map

The Birth of Modern Exploratory Data Analysis (1854)

- **Cholera outbreak** in London (Soho district).
- Snow mapped **cholera deaths geographically**.
- Identified **clustering** around a public water pump.
- Removing the pump handle helped control the outbreak.

<https://education.nationalgeographic.org/resource/mapping-a-london-epidemic/>

John Snow and the Cholera Map

This is one of the **earliest examples** of:

- **Data visualization** for insight
- Spatial analytics
- **Hypothesis generation** from data
- Public health analytics

Excellent prototype of EDA thinking.

John Snow and the Cholera Map



Data Collection and Mapping

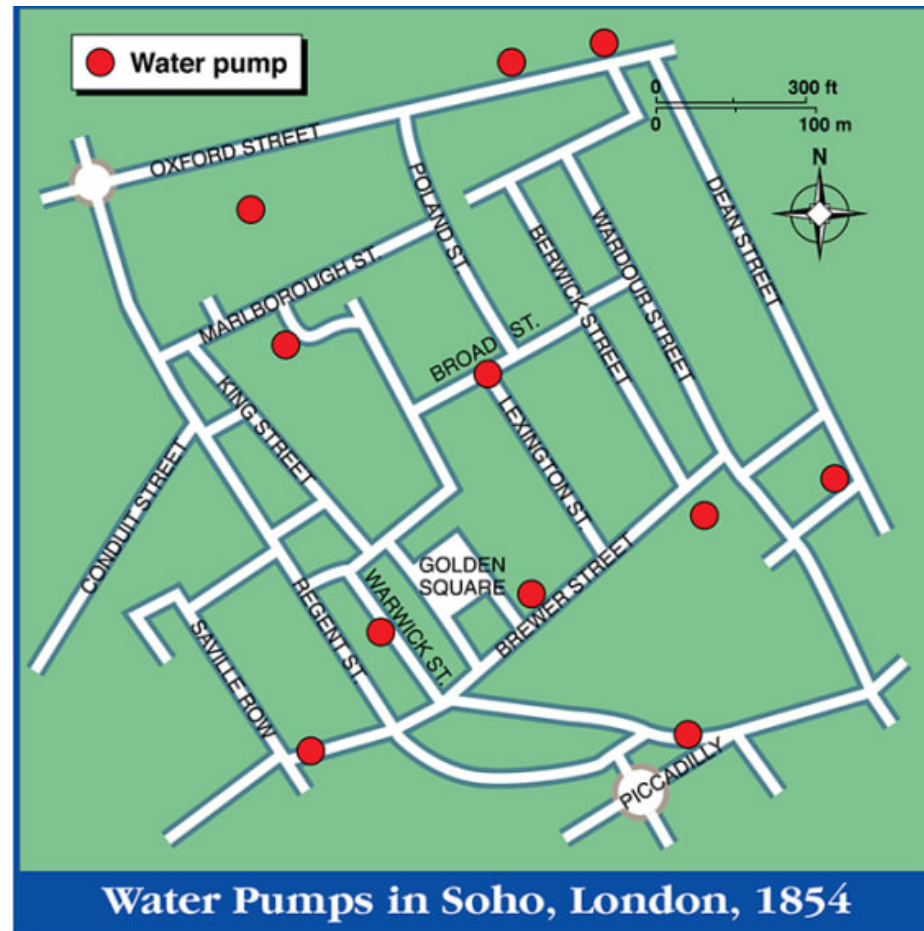
The Cholera Map Analysis

- Snow mapped **locations of cholera deaths**.
- Clear clustering around the Broad Street **water pump**.
- **Compared** affected vs unaffected households.
- **Spatial visualization** revealed a likely source.

Example of early:

- Exploratory Data Analysis (EDA)
- Data visualization for insight
- Spatial epidemiology

Water pump locations



Data Collection and Mapping

Why This Case Matters

- **Removal of the pump** handle helped contain the outbreak.
- Strong example of data-informed public health action.
- Demonstrates:
 - **Power of visualization**
 - Importance of hypothesis-driven analysis
 - Evidence challenging dominant beliefs

John Snow's pump



<https://livinglondonhistory.com/the-story-of-the-john-snow-pump-in-soho/>

Analytics in the Industrial Era (~1900)

Rise of Scientific Management

- **Industrialization increased data generation:**

- Production metrics
- Labor efficiency
- Quality **control**

- Frederick Taylor's scientific management:

- **Measurement** of work processes
- **Optimization using data**

Analytics became organizational and operational.

The Emergence of Computing

From **Manual** Analytics to **Digital** Analytics

Early 20th century developments:

- **Mechanical tabulating machines** (e.g., census processing).
- Later **electronic computers** (mid-20th century).
- Enabled:
 - Large-scale **statistical analysis**
 - Automation of calculations
 - Data storage and retrieval

Computers amplified analytics — they did not create it.

Transition to Modern Data Analytics

From Statistics to Data Science

- Expansion of **statistical** methods.
- **Databases** and **business intelligence** systems.
- **Machine learning** and **AI**.
- **Big Data** era.

Data-analytic thinking remains the **foundation**.

Final Message

What Students Should Remember

- Analytics **predates** computers.
- Data thinking evolved through trade, science, industry, and technology.
- **Visualization** and **observation** remain central (John Snow example).
- **Tools change — analytical thinking endures.**

Core idea:

EDA is the continuation of a centuries-old analytical mindset.

Obrigado!